EE2211-Midterm-AY2526-Sem1 - Confidential

# of Questions: 24
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Question #: 1

Consider the dataset with one dimensional features $\mathbf{x}_i \in \mathbb{R}$ and their targets $y_i \in \mathbb{R}$

$$\mathbf{x}_1 = 1 \quad \mathbf{x}_2 = 2 \quad \mathbf{x}_3 = 2.9$$
$$y_1 = 2 \quad y_2 = 4.1 \quad y_3 = 6.1$$

We would like to do linear regression and ridge regression in this problem.

(a) We would like to fit an affine model $y = w_0 + w_1\mathbf{x}$ using the method of least squares. Find w_0 and w_1 and write down your answers correct to 3 decimal places.

Your Answer to (a):

w_0 is 1 ;

w_1 is 2

(b) Now, we would like to use ridge regression with $\lambda = 0.1$. Again, find w_0 and w_1 and write down your answers correct to 3 decimal places.

Your Answer to (b):

w_0 is 3 ;

w_1

is 4

(c) Using the ridge regression model in part (b), we would like to predict the target value y_{new} of $\mathbf{x}_{\text{new}} = 1.5$. Find y_{new} and write your answers correct to 3 decimal places.

Your Answer to (c):

y_{new}

is 5 .

1. Range - Min:-0.176 Max:-0.174
2. Range - Min:2.156 Max:2.158
3. Range - Min:0.037 Max:0.039
4. Range - Min:2.047 Max:2.049
5. Range - Min:3.109 Max:3.111

Question #: 2

You are given a collection of 6 training data points of three features (x_1, x_2, x_3) and their one-hot encoding of class labels (y_1, y_2, y_3) which are packed as follows:

$$\mathbf{X} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 0 & 6 \\ 1 & 1 & 0 \\ 0 & 1 & 2 \\ 5 & 7 & -2 \\ -1 & 4 & 0 \end{bmatrix}, \mathbf{Y} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \text{class1} \\ \text{class1} \\ \text{class2} \\ \text{class3} \\ \text{class2} \\ \text{class3} \end{bmatrix}.$$

(i) Predict the class label of $(x_1, x_2, x_3) = (1, -2, 3)$ using the linear regression model with a bias term. My prediction for the class label is class 1 (enter 1, 2, or 3?).

(ii) Predict the class label of $(x_1, x_2, x_3) = (1, -2, 3)$ using the full polynomial model of third order. My prediction for the class label is class 2 (enter 1, 2, or 3?).

Please provide your answer below

1) 1

2) 2

1. Choice of: 1 | 2 | 3 - Correct Answer:2
2. Choice of: 1 | 2 | 3 - Correct Answer:3

Question #: 3

Which of the following statement(s) is/are true?

I. Given three samples of two-dimensional data points

$$\mathbf{X} = \begin{bmatrix} 1 & 4 \\ 5 & -1 \\ 2 & 3 \end{bmatrix}$$

with the corresponding target vector

$$\mathbf{y} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

. Suppose you want to use a full second-order polynomial model to fit these data, the polynomials model has 6 parameters to learn.

II. Given three samples of two-dimensional data points

$$\mathbf{X} = \begin{bmatrix} 1 & 4 \\ 5 & -1 \\ 2 & 3 \end{bmatrix}$$

with the corresponding target vector

$$\mathbf{y} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

. Suppose you want to use a full second-order polynomial model to fit these data, the polynomials model is an under-determined system.

III. Suppose we have 5 training samples, we would like to perform multi-class classification to predict y (4 classes) using input with 6 features using the optimal linear regression model with an offset term. In this case, the estimated parameter matrix **W** is of dimension 6 x 4.

IV. Suppose we have 5 training samples, we would like to perform multi-class classification to predict y (4 classes) using input with 6 features using the optimal linear regression model with an offset term. In this case, the estimated parameter matrix **W** is of dimension 4 x 6.

✓A. I &II
B. I &III
C. I &IV
D. I
E. None of the others

Question #: 4

The values of feature x and their corresponding values of target y are shown in the table below.

x	3	4	5	6	7
y	5	4.5	4	3.5	3

Based on the least square regression line y =a *x* +b, please select the correct option below.

I) The estimated a* and b* minimize the sum of squared errors between the predicted y' and the target output y for all training samples.

II) The sum of squared errors between the predicted y' and the target output y for all training samples is called the loss function.

III) The estimation of y at *x* =4.5 may be accurate as the prediction range is within the given data range.

IV) The estimation of y at *x* =1000 should be accurate as the prediction is linear.

✓A. I &III
B. III
C. I, II &III
D. I &IV
E. None of the above

Question #: 5

The solution for a ridge regression problem with feature vectors in **X** and targets in **y** can be written as

$$\widehat{\mathbf{w}} = (\mathbf{X}^T\mathbf{X}+\lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$$

for λ >0. Which of the following is/are true?

I. As λ increases,

$$\widehat{\mathbf{w}}^T\widehat{\mathbf{w}}$$

increases.

II. As λ increases,

$$\widehat{\mathbf{w}}^T\widehat{\mathbf{w}}$$

decreases.

III. As λ increases, there is more penalty on parameters.

IV. As λ increases, there is less penalty on parameters.

A. I &III
B. I &IV
✓C. II &III
D. II &IV
E. None of the above

Question #: 6

A set of linear equations is written as

$$\mathbf{w}^T\mathbf{X} = \boldsymbol{y}^T$$

where

$$\mathbf{X} \in \mathcal{R}^{4\times 3}$$

and

$$\mathbf{y} \in \mathcal{R}^{3\times 1}$$

. How many simultaneous equations are there in this set of equations?

A. 1
B. 2
✓C. 3
D. 4
E. 5

Question #: 7

Which of the following is/are true regarding derivatives?

I) Suppose g(**x**) is a *scalar* function of *b* variables where **x** is a *bx*1 vector. Without taking data points into consideration, the outcome of differentiation of g(**x**) w.r.t. **x** is a scalar.

II) Suppose f(**x**) is a *scalar* function of *d* variables where **x** is a *dx*1 vector. Without taking data points into consideration, the outcome of differentiation of f(**x**) w.r.t. **x** is a *dx1* vector.

III) If **f**(**x**) is a vector function of *size b*x1 and **x** is a *d*x1 vector, then differentiation of **f**(**x**) w.r.t. **x** is a *bxd* matrix.

IV) If **f**(**x**) is a vector function of *size b*x1 and **x** is a *d*x1 vector, then differentiation of **f**(**x**) w.r.t. **x** is a *dx1* vector.

A. I &III
B. I &IV
C. II &IV
✓D. II &III
E. None of the above

Question #: 8

A vector function has the mapping of

$$f(\mathbf{x}): \mathcal{R}^3 \rightarrow \mathcal{R}^2$$

. The gradient of this function with respect to **x** is a matrix of partial derivatives, which is a 2×3 matrix.

✓A. True
B. False

Question #: 9

This question is related to the determination of types of system where an appropriate solution can be found subsequently. The linear system below has no *exact* solution.

$$\begin{bmatrix} 1 & 3 & 2 \\ 3 & 6 & 4 \\ 5 & -1 & 6 \\ 9 & 8 & 0 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 9 \\ 3 \end{bmatrix}$$

✓A. True
B. False

Question #: 10

This question is related to determination of types of system where an appropriate solution can be found subsequently. The following matrix has a left inverse.

$$\mathbf{X} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

A. True
✓B. False

Question #: 11

The linear regression is the special case of polynomial regression of order 1.

✓A. True
B. False

Question #: 12

A learning problem is attempted by minimizing the sum of squared errors given by

$$\sum_{i=1}^{m} (f_{\mathbf{w},b}(\mathbf{x}_i) - y_i)^2 = (\mathbf{X}\mathbf{w} - \mathbf{y})^T (\mathbf{X}\mathbf{w} - \mathbf{y}) \quad \text{where} \quad \mathbf{X} \in \mathcal{R}^{10\times 11} \quad \text{and} \quad \mathbf{w} \in \mathcal{R}^{11\times 1}$$

denote respectively the data matrix and the parameter vector. The outcome must be a unique solution since the square function is convex.

A. True
✓B. False

Question #: 13

Data Wrangling is one typical step of data validation.

A. True
✓B. False

Question #: 14

For a random experiment, both outcomes and events are not decomposable.

A. True
✓B. False

Question #: 15

The Bayes' Rule holds only when X and Y are independent random variables.

A. True
✓B. False

Question #: 16

In a 2D coordinate system, any pair of vectors is linearly dependent.

A. True
✓B. False

Question #: 17

Simpson's paradox is a common way for extracting features for classification.

A. True
✓B. False

Question #: 18

We are given the set of average daily temperatures of Singapore in the past 20 years (i.e., each sample in the set denotes the average temperature of one day), and we are asked to group them into three groups without any supervision. This task, by nature, is a classification task.

A. True
✓B. False

Question #: 19

Machine learning relies on using reinforcement learning for clustering.

A. True
✓B. False

Question #: 20

Please select the correct option.

i) Simpson's paradox is a serious issue, but it can be solved using data visualization.

ii) In 2D space, a vector can be visualized as an arrow or a point; however, in 3D space, a vector can be visualized only as a point.

iii) One solution to identify causality between two variables is to check their c orrelation. If they are correlated, they are known to have causality.

iv) For any classifier, input data (i.e., extracted features) and output data (i.e., predictions) must be of the same data type.

A. i)
B. ii)
C. iii)
D. iv)
E. iii), and iv)
F. i), and iii)
G. ii) and iii)
✓H. None of the others is correct.

Question #: 21

Please select the correct option.

i) Binary coding applies to interval data, but not to nominal data.

ii) Feature extraction applies to nominal data, but not to interval data.

iii) Data cleaning applies to ratio data, but not to ordinal data.

iv) Data visualization applies to ratio data, but not to ordinal data.

A. i)
B. ii)
C. iii)
D. iv)
E. i), ii), and iv)
F. i) and iv)
G. i) and ii)
H. ii), iii), and iv)
✓I. None of others is correct.

Question #: 22

We are designing a machine learning model to classify hand gestures captured by a camera into one of the following categories: "Thumbs Up", "OK", "Stop", "Wave", and "Fist".

We have collected a dataset of images, containing 400 images for each category (a total of 2,000 images). We then ask an expert to review all 2,000 images. The expert eventually identifies 200 poor-quality images and suggests that we remove these images from the dataset. We follow the suggestion and end up with 1,800 images.

Which of the following statement(s) is/are correct?

i) The expert inspection process can be considered a step for feature extraction.

ii) Given a new image, the system predicts one of the five categories. This can be considered a regression task.

iii) The category of the image (i.e., "Thumbs Up", "OK", "Stop", "Wave", and "Fist") belongs to ordinal data.

iv) One possible one-hot encoding for the five categories is as follows: [0 0 0 0 0], [0 0 0 1 1], [0 1 1 0 0], [1 1 1 1 0], [1 0 1 0 1].

v) Given a new image, the system predicts one of the five categories. This can be considered a clustering task, since we are grouping the new image to the existing categories.

A. i)
B. ii)
C. iii)
D. iv)
E. v)
F. i), ii), and iii)
G. iv) and v)
H. i), ii), iii), and iv)
✓I. None of the others.

Question #: 23

A technology company is developing AutoDrive, an intelligent AI system for autonomous vehicles. AutoDrive integrates camera images, LiDAR signals, GPS coordinates, and driver behavior logs to enable safe and adaptive driving in complex real-world environments. You are part of the AI research team designing the machine learning pipeline.

AutoDrive uses a dataset of camera images labeled as “Pedestrian,” “Vehicle,” “Traffic Light,” and “Road Sign.” The goal is to train a model for predicting a new image into ones of these classes. What learning paradigm should be used for this perception task?

A) Supervised Learning
B) Unsupervised Learning
C) Reinforcement Learning

Your answer: __1__

To automatically detect unusual driving behaviors (e.g., sudden lane changes) without labeled data, AutoDrive employs anomaly detection. This is an example of:

A) Supervised Learning
B) Unsupervised Learning
C) Reinforcement Learning

Your answer: __2__

In AutoDrive, road surface quality is rated as “Poor,” “Fair,” “Good,” or “Excellent.” This variable represents:

A) Nominal Data
B) Ordinal Data
C) Interval Data
D) Ratio Data

Your answer: 3

The feature “Vehicle Speed (km/h)” is continuous and measured from 0 to 200. To bring it to a comparable scale between 0 and 1, the team applies:

A) One-Hot Encoding
B) Linear Scaling
C) Data Wrangling

Your answer: 4

1. Choice of: A) | B) | C) - Correct Answer:A)
2. Choice of: A) | B) | C) - Correct Answer:B)
3. Choice of: A) | B) | C) | D) - Correct Answer:B)
4. Choice of: A) | B) | C) - Correct Answer:B)

Question #: 24

A company has 800 employees divided into three departments: Department A has 300 employees, Department B has 200 employees, Department C has 300 employees.

•Department A consists of two teams: Team A-1: 200 employees, and Team A-2: 100 employees.
•Department B consists of 1 team only, Team B with 200 employees.
•Department C consists of two teams: Team C-1: 150 employees, and Team C-2: 150 employees.

At the end of the year, the company conducts a performance review where each employee either gets promoted ('p') or not ('n').

Promotion Results is as follows:

In Department A: 200 employees get promoted, among whom 150 are from Team A-1.
In Department B: 80 employees get promoted.
In Department C: 150 employees get promoted, among whom 90 are from

Team C-1.

1. What is the probability that a randomly selected employee gets promoted, i.e., P(X='p')?

Your Answer __1__ (Write your answer to three decimal places, between 0 and 1.)

2. Given that an employee was not promoted, what is the probability that they are NOT from Team B, i.e., P(T != 'B' | X='n')?

Your Answer __2__ (Write your answer to three decimal places, between 0 and 1.)

3. Which team has the highest promotion rate?

Your Answer __3__

4. Which department has the highest promotion rate?

Your Answer __4__

1. Range - Min:0.53 Max:0.54
2. Range - Min:0.67 Max:0.68
3. Choice of: Team A-1 | Team A-2 | Team B | Team C-1 | Team C-2 - Correct Answer:Team A-1
4. Choice of: Department A | Department B | Department C - Correct Answer:Department A